An Abundance-Guided Attention Network for Hyperspectral Unmixing

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***Abstract*— Hyperspectral unmixing is a vibrant research field that focuses on the task of decomposing mixed pixels into a collection of pure spectral signatures, known as endmembers, along with their corresponding fractional abundances. Conventional unmixing algorithms often need to combine two techniques, namely endmember extraction and abundance estimation, to accomplish the unmixing task. Recently, deep learning (DL) has succeeded in the field of hyperspectral unmixing due to its strong feature learning and data-fitting capabilities. By extracting the output and weight of a particular layer as abundance maps and endmember signatures, available DL methods can directly unmix hyperspectral images. However, to improve the performance of spectral unmixing, such available DL methods frequently employ the results of endmember extraction algorithms—in most cases, the well-known vertex component analysis (VCA)—as the initial weights, which leads to significant limitations in their performance: 1) the unmixing results are heavily dependent on the initialization given by VCA and 2) the randomness of VCA is passed to the unmixing network. In this article, we design a new method called abundance-guided spectral and spatial network (A2SN) that not only skips the weights to extract endmember features directly from the network, but also estimates the abundance maps and reconstructs images directly. In particular, the proposed A2SN employs different kernels to capture spectral and spatial information. We also propose an abundance-guided attention spectral and spatial attention network (A2SAN) for hyperspectral unmixing by integrating attention mechanisms into A2SN. As a result, A2SAN is a completely innovative unmixing method that employs attention and reconstruction directly for hyperspectral unmixing, rather than just as modules for information extraction. Most importantly, both A2SN and A2SAN use a weighted summation of the feature maps to reconstruct the image and increase the noise immunity of the network. Experimental results, conducted on both synthetic and real datasets, demonstrate the effectiveness and superiority of A2SN and A2SAN over state-of-the-art unmixing methods. Our full code is released at https://github.com/xuanwentao/A2SN-and-A2SAN for public evaluation.**

***Index Terms*— Abundances, attention, endmembers, hyperspectral images, spectral and spatial information, unmixing.**

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I. Introduction

HYPERSPECTRAL images [1], [2], [3] contain rich and detailed spatial and spectral information and have been applied in different tasks, for example, change detection [4] and image classification [5]. However, given the inherent limitations of hyperspectral imaging devices in terms of spatial resolution, coupled with the intricate composition of materials in the observed scene, hyperspectral images frequently exhibit an important number of mixed pixels. Hyperspectral unmixing [6], [7], [8], [9], [10], as an excellent tool for analyzing hyperspectral images, is used to decompose mixed pixels into a set of pure spectral signatures (endmembers) and their corresponding abundances at each pixel. Traditional unmixing methods are mainly categorized into those based on the linear mixture model (LMM) [11], [12] and those based on the nonlinear mixture model (NLMM) [13], [14]. The LMM assumes that the observed pixel spectrum is a linear combination of endmember signatures and the corresponding weights are the abundance maps that need to be estimated. Considering that the LMM is simple and effective, hyperspectral unmixing techniques based on the LMM have been extensively developed in the literature.

A. Traditional Unmixing Methods

Hyperspectral unmixing encompasses three primary tasks, that is, determining the number of endmembers, extracting spectral signatures, and estimating their abundances for each pixel. The number of endmembers is typically treated as prior knowledge or a prerequisite for the extraction process. Various algorithms exist for determining the number of endmembers, falling into three main categories: information theory-based algorithms, eigenvalue thresholding algorithms, and geometry characterization algorithms. Information theory-based algorithms [15], [16], [17] depend on the empirical configuration

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of a likelihood function or particular mixture model. In this scenario, an improper configuration will lead to errors in the estimation of the number of endmembers, which will affect the subsequent endmember extraction and abundance estimation tasks. Eigenvalue thresholding algorithms usually apply thresholding to the eigendecomposition results of subspace analysis. Algorithms in this category include the hyperspectral signal subspace by minimum error (Hysime) [18], principal component analysis (PCA) [19], and Harsanyi–Farrand–Chang (HFC). Geometry characterization algorithms encompass the geometry-based estimation of the number of endmembers (GENEs) [20], along with its variants such as the convex hull-based (GENE-CH) algorithm and the affine hull-based (GENE-AH) algorithm.

Once the number of endmembers has been determined, a plethora of algorithms have been developed for extracting spectral signatures. Classical endmember extraction algorithms include vertex component analysis (VCA) [21], minimum volume simplex analysis (MVSA) [22], simplex growing algorithm (SGA) [23], among many others [24], [25], [26], [27]. VCA employs an iterative approach to project the data in a direction orthogonal to the subspace spanned by the already determined endmembers. The new endmembers are then identified as the pixels located at the edge of the projected data. MVSA utilizes a minimum volume simplex fitting technique, imposing the constraint that the abundance fractions should belong to the probability simplex. SGA identifies the convex geometry of endmembers with minimum volume, but it requires a specific initialization. It is important to note that all of the aforementioned endmember extraction algorithms necessitate prior knowledge of the number of endmembers. In this sense, some methods that do not require the number of endmembers, including the divergent subset (DS) [24] or the maximum distance analysis (MDA) [28], have been developed. In DS, a subset of pixels that exhibit the highest dissimilarity from each other is selected as the endmember set. This strategy ensures that the inclusion of additional pixels increases the similarity within the DS, thereby reducing its divergence. The number of endmembers is determined by the cardinality of the subset. MDA initially identifies the pixel point that is farthest from any other pixel point as the first endmember. Subsequently, it identifies the farthest pixels from any pixel, line, plane, and AH formed by all previously extracted endmembers as subsequent endmembers. The endmember extraction process in MDA concludes when the maximum distance between all pixel points and the AH formed by all extracted endmembers becomes zero.

Once the endmember signatures have been extracted, the subsequent crucial step involves estimating their abundance maps at each pixel. Representative abundance estimation algorithms include the fully constrained least-squares (FCLSs) [29], sparse unmixing by variable splitting and augmented Lagrangian (SUnSAL) [30], and spatial group sparsity regularized nonnegative matrix factorization (SGSNMF) [31]. SUnSAL uses the alternating direction method of multipliers (ADMMs) to decompose a difficult problem into a sequence of simpler ones. SGSNMF incorporates a constraint, namely, a spatial group sparsity regularizer, into nonnegative matrix factorization (NMF)-based methods to perform spectral unmixing. However, the aforementioned abundance estimation algorithms require endmember extraction results obtained from other techniques, for example, VCA. In this sense, the minimum-volume enclosing simplex (MVES) [32] and the robust collaborative NMF (R-CoNMF) [33] were developed to simultaneously complete the two tasks, that is, endmember extraction and abundance estimation. MVES incorporates convex analysis and Craig's criterion to enhance the performance of spectral unmixing. R-CoNMF determines the number of endmembers by starting with an overestimated number of endmembers and removing redundant endmembers through cooperative regularization.

B. DL-Based Unmixing Methods

In recent years, hyperspectral unmixing methods based on deep learning (DL) have been developed in the literature because DL exhibits powerful learning and data-fitting abilities [34], [35], [36], [37]. Existing DL-based unmixing methods [38], [39], [40], [41], [42] mainly include the untied denoising autoencoder (uDAS) [43], stacked nonnegative sparse autoencoder (NNSAE) [44], cycle-consistency unmixing network (CyCU-Net) [45], CNN autoencoder unmixing (CNNAEU) [46], spectral convolution network with a spectral library (SCNL) [47], and convolutional autoencoder unmixing (CAU) [48]. uDAS primarily tackles the challenge of unsupervised unmixing in the presence of high noise. It incorporates denoising capabilities by imposing a constraint that minimizes reconstruction errors (REs). Additionally, uDAS employs matrix operations to efficiently ensure nonnegativity and sum-to-one constraints on the endmembers. NNSAE can handle outliers and the low signal-to-noise ratio (SNR) in the data. In particular, NNSAE improves the performance of unmixing by considering the sparsity and nonnegative properties of the abundance vectors. CyCU-Net can reduce the loss of material-related information during the reconstruction process by relaxing the original pixel-level reconstruction assumptions to cycle consistency dominated by a cascaded AE. CNNAEU utilizes the spatial and spectral structure of the hyperspectral image for endmember extraction and abundance estimation purposes. Especially, CNNAEU employs 3-D patches and does not use any pooling or upsampling layers to preserve the spatial structure, and the feature maps for a hidden convolutional layer are the abundance maps that need to be estimated. SCNL exploits a spectral library to improve the representation capabilities of the network, and CAU uses some convolutional layers (followed by an autoencoder) to perform spectral unmixing, and it can learn spatial distributions in the image via convolutional filters, which are then applied to parameterize innate spectral features. However, CAU is designed based on the LMM and may not provide accurate results for NLMM problems. It should be noted that all of these methods estimate endmembers and abundances from the weight and output of a specific layer and use VCA to initialize the weights. Thus, they achieve good performance only when the initial VCA results are correct.

Very recently, a new model called vision transformer (ViT) [49], [50] has been proposed that can capture nonlocal

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contextual feature dependencies. Different from CNNs, the transformer mainly exploits self-attention (which is the key of the transformer) to analyze hyperspectral images more effectively and encode the pixel positions to capture global information. In [51], a novel attention-based residual network with scattering transform (ARNST) features is proposed for hyperspectral unmixing. ARNST combines the scattering transform and a deep neural network (e.g., a CNN) to extract deep-level features and process the resulting high-order information, respectively. In [52], an adaptive spectral and spatial attention autoencoder network (SSANet) is proposed for spectral unmixing. The corresponding spectral attention module learns the weights of spectral bands in the input data to enhance the representation of spectral information, and the spatial module obtains the attention weight assigned to each adjacent pixel to derive useful spatial information. In [53], a convolutional autoencoder (CAE) and a transformer are combined to estimate endmember signatures and abundances effectively. However, like existing DL-based unmixing methods, transformer-based methods also extract the weights as endmembers and the output of a layer as abundance maps.

C. Motivation and Contributions

After reviewing traditional unmixing methods in Section I-A, we conclude that most of them are designed for two independent tasks, that is, endmember extraction and abundance estimation, which means that the search for a combination of two good algorithms for the overall unmixing task is necessary and costly. Moreover, the final performance of abundance estimation will be constrained by the initial choice made for endmember extraction.

After reviewing DL-based unmixing methods in Section I-B, it is observed that they are mainly designed as end-to-end models where the input is the hyperspectral image and the corresponding output is the reconstructed hyperspectral image. They estimate abundance maps from the output of a specific layer and the number of nodes of this layer must be set as the number of endmembers. The endmember signatures are generally regarded as the weights between the specific layer (i.e., the one for estimating abundances) and the reconstructed layer. Especially, these unmixing networks based on DL normally use the endmember extraction results by VCA as initial weights. In this sense, how to extract endmembers directly from the network (rather than through the weights) deserves further investigation.

To overcome the aforementioned challenges faced by traditional and DL-based unmixing methods, a new abundance-guided spectral and spatial network (A2SN) is proposed for hyperspectral unmixing that extracts endmember signatures by calculating the product of the feature map and the abundance map, estimates abundances as the output of a specific layer, reconstructs images as the product of the feature and abundance matrix and its transpose. In this sense, the proposed A2SN is not only capable of directly extracting endmembers without the weights of the corresponding network, but it also simultaneously estimates abundances and reconstructs the input images. Moreover, a self-attention mechanism is embedded into A2SN to estimate endmembers and abundances directly, instead of functioning only as a module of the network. More specifically, the innovative contributions of this work can be summarized as follows.

1) A novel unmixing method called A2SN is proposed for simultaneously extracting endmembers, estimating abundance maps, and reconstructing images. To the best of our knowledge, this is the first attempt in the literature to extract endmembers by the product of the feature map and abundance matrix (instead of weights), thus avoiding the random initialization required by other methods.
2) A self-attention mechanism is embedded into A2SN to propose a new network called attention spectral and spatial attention network (A2SAN) for hyperspectral unmixing. This represents the first attempt in the literature to use attention to directly estimate endmembers and abundance maps and to reconstruct images, rather than using it just as a processing module for capturing information (e.g., extracting deep-level features), as in other networks.
3) In the proposed A2SN and A2SAN, an abundance-guided mechanism is incorporated innovatively, attempting to use the product of the abundance and its transpose to measure the similarity of the feature maps to reconstruct hyperspectral images. Most importantly, the proposed A2SN and A2SAN weight feature maps by similarity to obtain the final reconstructed images and exhibit robustness to noise.

D. Paper Organization

We structure the remainder of this article as follows. Section II provides an overview of related works. In Section III, we detail the proposed A2SN and A2SAN methods. Section IV presents the experimental datasets, along with the settings, results, and corresponding discussions. Finally, Section V concludes this article, offering remarks and suggesting potential avenues for future research.

II. Related Work

The newly proposed A2SN and A2SAN are designed based on the LMM and the DenseNet, and the attention mechanism in A2SAN is used to directly estimate abundance maps. In this section, the LMM, the attention mechanism, and DenseNet are reviewed.

A. LMM

Let $\mathbf{Y} \in \mathbb{R}^{Nb\times Np}$ denote the hyperspectral data, where Nb is the number of spectral bands and Np is the number of pixels. Under the LMM, the observed spectral reflectance is formulated as

$$\mathbf{Y} = \mathbf{MA} + \mathbf{N} \tag{1}$$

where $\mathbf{M} \in \mathbb{R}^{Nb\times c}$ denotes the endmember signatures, $\mathbf{A} \in \mathbb{R}^{c\times Np}$ represents the abundance maps, $\mathbf{N} \in \mathbb{R}^{Nb\times Np}$ is noise, and c is the number of endmembers. Abundance maps need to satisfy the abundance nonnegative constraint (ANC) and abundance sum-to-one constraint (ASC), that is,

$$\text{ANC}: a_{ij} \geq 0 \quad \forall i, j \tag{2}$$

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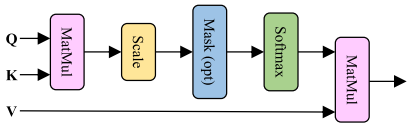


Fig. 1. Structure of the self-attention mechanism.

and

$$\text{ASC}: \sum_{i=1}^{c} a_{ij} = 1 \quad \forall i, j. \tag{3}$$

B. Self-Attention and Efficient Attention Mechanisms

1) Self-Attention Mechanism: The self-attention mechanism [54] is used to learn long-distance dependency relations of the input data and plays a key role in image processing. Fig. 1 shows the structure of the self-attention mechanism. More specifically, its main steps include as follows.

1) Input the sequence data $\tilde{\mathbf{Y}} \in \mathbb{R}^{L\times d}$, where L is the length and d is the dimension.
2) Project $\tilde{\mathbf{Y}}$ into three matrices.
   a) *Query:* $\mathbf{Q} = \mathbf{W}^{\mathrm{Q}}\tilde{\mathbf{Y}}$.
   b) *Key:* $\mathbf{K} = \mathbf{W}^{\mathrm{K}}\tilde{\mathbf{Y}}$.
   c) *Value:* $\mathbf{V} = \mathbf{W}^{\mathrm{V}}\tilde{\mathbf{Y}}$.
3) Compute the attention score between $\mathbf{Q}$ and $\mathbf{K}$ by

$$\mathbf{S} = \mathbf{Q}\mathbf{K}^{\mathrm{T}} \tag{4}$$

   where T denotes the transpose. Especially, to stabilize the gradients, $\mathbf{S}$ usually performs the normalization and updates to $\mathbf{S} = (\mathbf{Q}\mathbf{K}^{\mathrm{T}}/\sqrt{d})$, but this operation is not necessary.
4) Apply the softmax function to $\mathbf{S}$ on position-1 and update $\mathbf{S}$ to

$$\mathbf{S} = \text{softmax}(\mathbf{Q}\mathbf{K}^{\mathrm{T}}). \tag{5}$$

5) Obtain the output by

$$\mathbf{Z} = \mathbf{S}\mathbf{V}. \tag{6}$$

To sum up, the self-attention mechanism can be integrally formulated as follows:

$$\mathbf{Z} = \text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}(\mathbf{Q}\mathbf{K}^{\mathrm{T}})\mathbf{V}. \tag{7}$$

2) Efficient Attention Mechanism: The original self-attention mechanism involves dot products, but its corresponding memory and computation costs grow quadratically with the input size. To address the issue, Shen et al. [55] proposed an efficient attention mechanism that is equivalent to the original self-attention mechanism but with much lower memory and computational requirements. In such a mechanism, the attention score is obtained by

$$\mathbf{S} = \text{softmax}(\mathbf{Q})\text{softmax}(\mathbf{K})^{\mathrm{T}} \tag{8}$$

and the final output is obtained by

$$\mathbf{Z} = \text{softmax}(\mathbf{Q})(\text{softmax}(\mathbf{K})^{\mathrm{T}}\mathbf{V}). \tag{9}$$

It is easy to find that the difference between the traditional and the efficient attention mechanisms mainly comprises the following aspects.

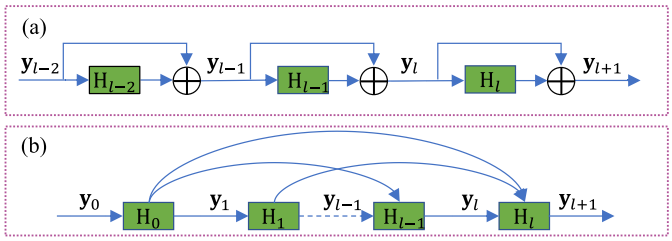


Fig. 2. Structure of (a) ResNet and (b) DenseNet.

1) The efficient attention mechanism performs softmax on $\mathbf{Q}$ and $\mathbf{K}^{\mathrm{T}}$ instead of softmax on $\mathbf{Q}\mathbf{K}^{\mathrm{T}}$.
2) The efficient attention mechanism changes the order of multiplication. Specifically, it uses $\mathbf{Q}(\mathbf{K}^{\mathrm{T}}\mathbf{V})$ to replace $\mathbf{Q}\mathbf{K}^{\mathrm{T}}\mathbf{V}$. This operation reduces the computational complexity from $O(n^2)$ to $O(d_k d_v)$, where d_k and d_v are the dimensions of $\mathbf{K}$ and $\mathbf{V}$, respectively.

C. ResNet and DenseNet

In general, more convolution layers will result in better performance, but many layers will result in problems of gradient disappearance and explosion. To address this issue, ResNet [56] uses skip connections to enable the input data to pass directly through the network. A simple structure of ResNet is illustrated in Fig. 2(a). It is easy to see that ResNet is composed of its basic unit (residual block), where the output $\mathbf{y}_l$ of the lth residual block can be computed by

$$\mathbf{y}_l = H_l(\mathbf{y}_{l-1}) + \mathbf{y}_{l-1} \tag{10}$$

where $H_l(\cdot)$ is a hidden block that contains convolutional layers, activation layers, and batch normalization layers.

Compared with ResNet, DenseNet [57] connects all layers by concatenation in the channel dimension to ensure maximum information flow between each layer of the network, and the dense block is its basic unit. Fig. 2(b) illustrates the structure of DenseNet. Here, the output $\mathbf{y}_l$ of the lth dense block is computed by

$$\mathbf{y}_l = H_l(\mathbf{y}_0, \mathbf{y}_1, \ldots, \mathbf{y}_{l-1}) \tag{11}$$

where H_l includes convolutional layers, activation layers, and batch normalization layers, and $\mathbf{y}_0, \mathbf{y}_1, \ldots \mathbf{y}_{l-1}$ are the feature maps generated by the previous dense blocks.

III. Proposed Methods

In this section, we present the proposed A2SN and A2SAN frameworks and their main modules, that is, spectral and spatial modules, abundance-guided unmixing module, abundance-guided attention unmixing module, and loss function.

A. Spectral and Spatial Modules

2D-CNNs are normally used to capture spatial information and 3D-CNNs are employed to capture spectral and spatial information in unmixing networks. In our work, we consider 2D-CNNs with different kernel sizes to learn the spectral and spatial information contained in hyperspectral images.

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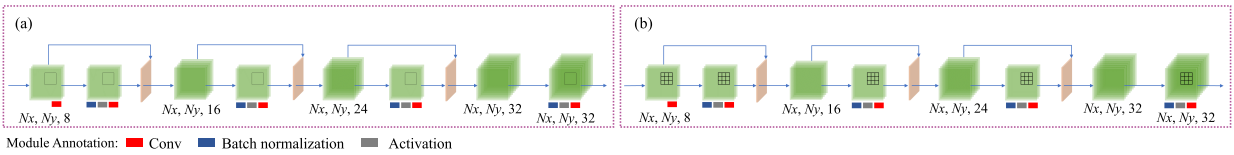


Fig. 3. Proposed spectral and spatial modules. (a) Spectral module (kernel size: 1×1, kernel number: 8). (b) Spatial module (kernel size: 3×3, kernel number: 8).

The structure of the spectral and spatial modules in our proposed A2SN and A2SAN is shown in Fig. 3. In the spectral module, we consider a kernel size of 1×1 for convolutional layers to capture spectral information. In our spectral module, because the kernel size of 1×1 does not consider the relationship between pixels and adjacent pixels in the spatial field, the convolution will not learn any spatial features. Moreover, the kernel size of 1×1 can linearly combine or integrate spatial information for each pixel in the spatial field. In this sense, the spectral information will be learned and the spatial information will be retained perfectly. In the spatial module, we use the same structure as in the spectral module but employ a kernel size of 3×3 for convolutional layers (instead of a kernel size of 1×1) to learn spatial information.

More specifically, in the proposed A2SN and A2SAN, we use the mechanism of the DenseNet introduced in Section II-C to construct the spectral and spatial modules. We first convolve the input image with a kernel number of 8 and a kernel size of 1×1 for the spectral module and a kernel size of 3×3 for the spatial module and then add batch normalization and the rectified linear unit (ReLU) activation function before each of the subsequent convolution layers with the same kernel size and number. Furthermore, we merge all outputs and input, and the feature maps ($8 \times 4 = 32$)—with the same spatial size as the input image—are generated. Finally, we perform batch normalization and ReLU for the 32 feature maps and convolve them using a kernel number of 32 and a kernel size of 1×1 for the spectral module and a kernel size of 3×3 for the spatial module to obtain their respective final outputs.

B. Abundance-Guided Unmixing Module

Normally, the similarity between two pixels in a hyperspectral image is computed by

$$\cos\theta = \frac{\mathbf{y}_i \cdot \mathbf{y}_j}{||\mathbf{y}_i||_2||\mathbf{y}_j||_2} = \frac{\mathbf{y}_i \cdot \mathbf{y}_j}{\sqrt{\sum_{k=1}^{L} {y_{ki}}^2}\sqrt{\sum_{k=1}^{L} {y_{kj}}^2}} \tag{12}$$

where the symbol $\cdot$ denotes the dot product, θ is the angle between $\mathbf{y}_i$ and $\mathbf{y}_j$, and $\mathbf{y}_i$ and $\mathbf{y}_j$ are more similar as the value of $\cos\theta$ is larger. Using (12) to find the similarity of pixels involves a lot of operations. Here, we directly exploit the dot product of $\mathbf{y}_i$ and $\mathbf{y}_j$ to measure their similarity [58], [59], [60], and the similarity between pixels in a hyperspectral image is computed by

$$W_{ij} = \mathbf{y}_i \cdot \mathbf{y}_j = ||\mathbf{y}_i||_2||\mathbf{y}_j||_2 \cos\theta. \tag{13}$$

By observing (12) and (13), it is not difficult to find that (12) is concerned with angular differences while (13) is concerned with angular and length differences. Therefore, we use the dot product to measure the similarity, and the similarity matrix of different pixels is obtained by

$$\mathbf{W} = \mathbf{Y}^{\mathrm{T}}\mathbf{Y} \tag{14}$$

where $\mathbf{W} \in \mathbb{R}^{Np \times Np}$, and the ith row or column represents the similarity of the ith pixel with other all pixels.

In Section II-A, we noticed that hyperspectral images can be denoted $\mathbf{Y} = \mathbf{MA} + \mathbf{N}$ under the LMM. Without considering the noise, the hyperspectral image can be expressed as

$$\mathbf{Y} = \mathbf{MA}. \tag{15}$$

That is, each pixel in a hyperspectral image can be formed by mixing the endmember matrix at a certain ratio (abundance maps). In the abundance matrix $\mathbf{A} \in \mathbb{R}^{c \times Np}$, the ith row represents the proportion of the ith endmember in each pixel of the hyperspectral image, and the ith column represents the proportion of each endmember in the ith pixel of the image. Two pixels will be more similar when the percentages of each endmember in the pixels are also more similar. Thus, the product of the abundance matrix can also reflect the similarity between pixels. In this sense, the similarity matrix is updated to

$$\mathbf{W} = \hat{\mathbf{A}}^{\mathrm{T}}\hat{\mathbf{A}} \tag{16}$$

where $\mathbf{W} \in \mathbb{R}^{Np \times Np}$, $\hat{\mathbf{A}}$ is the estimated abundance matrix, and $\hat{\mathbf{A}}^{\mathrm{T}}$ is the transpose of $\hat{\mathbf{A}}$. The final reconstructed image is obtained by

$$\hat{\mathbf{Y}} = \tilde{\mathbf{Y}}\mathbf{W} = \tilde{\mathbf{Y}}\hat{\mathbf{A}}^{\mathrm{T}}\hat{\mathbf{A}} \tag{17}$$

where $\tilde{\mathbf{Y}} \in \mathbb{R}^{Nb \times Np}$ represents the reconstructed image obtained by the spectral and spatial modules. Using (17) to obtain the final reconstructed image is equal to weighting each pixel, where the weight (i.e., the similarity) reflects the degree of contribution of different pixels to that pixel. With the similarity matrix, the original image can be reconstructed accurately. Moreover, using (17) to obtain the final image means that the pixels with noise will be weighted by other pixels, which increases the noise immunity of the network. Observing (15) and (17), the endmember matrix can be obtained by

$$\hat{\mathbf{M}} = \tilde{\mathbf{Y}}\hat{\mathbf{A}}^{\mathrm{T}}. \tag{18}$$

Thus, (17) is used to design a novel unmixing network, called A2SN, for extracting endmembers, estimating abundances, and reconstructing hyperspectral images simultaneously.

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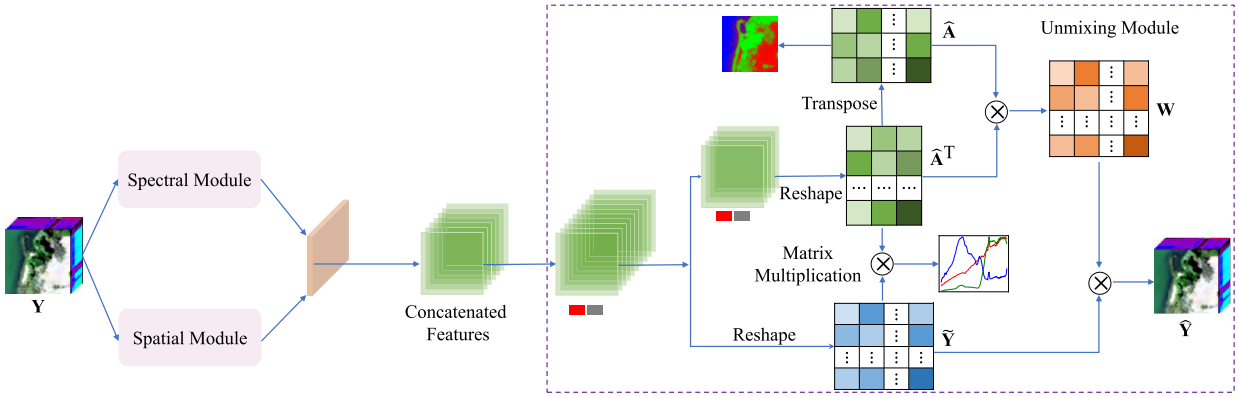


Fig. 4. Architecture of the proposed A2SN includes spectral and spatial modules for capturing information and an unmixing module for extracting endmembers, estimating abundance maps, and reconstructing images.

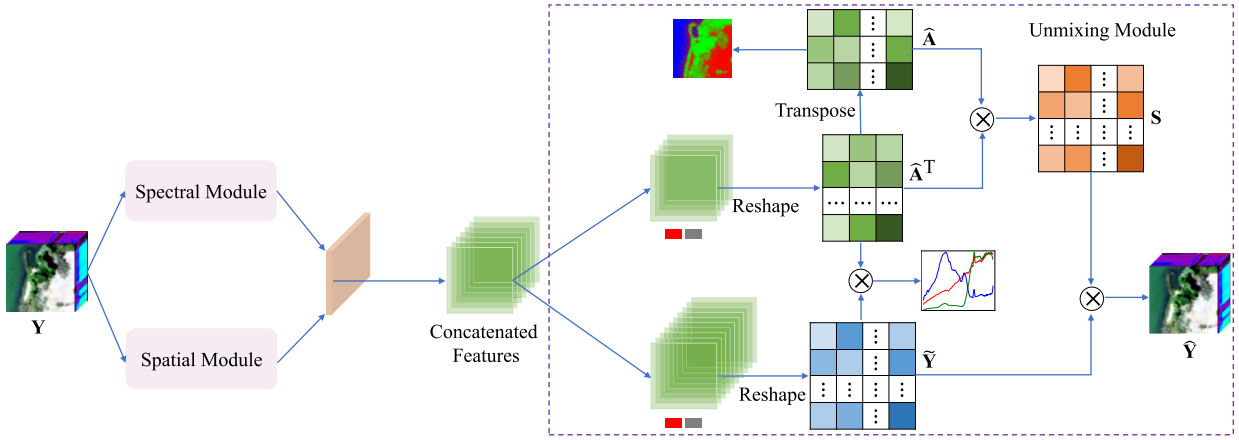


Fig. 5. Architecture of the proposed A2SAN for hyperspectral unmixing and reconstruction.

Fig. 4 shows the architecture of the proposed A2SN. Features obtained by the spectral and spatial modules are concatenated and the reconstructed image $\tilde{\mathbf{Y}}$ is obtained by convolving the concatenated features using a 2D-CNN with a kernel size of 1×1 and a kernel number of L features. Furthermore, a 2D-CNN with a kernel size of 1×1 and a kernel number of c is used to convolve the reconstructed image $\tilde{\mathbf{Y}}$ and obtain the abundance maps. The reshape operation is used to obtain $\hat{\mathbf{A}}^{\mathrm{T}}$, and the transpose operation is employed to obtain $\hat{\mathbf{A}}$. Then, the reconstructed image and endmember signatures are obtained by (17) and (18), respectively.

C. Abundance-Guided Attention Unmixing Module

Observing the process of A2SN, it is found that the unmixing module is very similar to the self-attention mechanism. Therefore, a standard self-attention mechanism is embedded in A2SN to introduce A2SAN for hyperspectral unmixing and explain the abundance-guided process from the viewpoint of the attention mechanism.

The architecture of A2SAN is presented in Fig. 5. As we can see, it has the same spectral and spatial modules as A2SN. However, different from A2SN, the abundance maps and reconstructed images are obtained by convolving the concatenated features using different kernel numbers c and L, respectively. Then, we embed the attention mechanism in the unmixing module, and the attention score is obtained by

$$\mathbf{S} = \hat{\mathbf{A}}^{\mathrm{T}}\hat{\mathbf{A}} \tag{19}$$

where $\hat{\mathbf{A}}$ and $\hat{\mathbf{A}}^{\mathrm{T}}$ play the roles of softmax $\mathbf{K}^{\mathrm{T}}$ and softmax $\mathbf{Q}$, respectively. Especially, the abundance matrix $\hat{\mathbf{A}}$ meets the ANC and ASC constraints, and the additional softmax does not need to be performed on $\hat{\mathbf{A}}$ and $\hat{\mathbf{A}}^{\mathrm{T}}$. By reviewing the efficient attention mechanism introduced in Section II-B2, we know that changing the order of multiplications not only does not affect the final result but also reduces computational overload. Therefore, the output of A2SAN is obtained by

$$\begin{aligned}\hat{\mathbf{Y}} &= \tilde{\mathbf{Y}}\mathbf{S} \\ &= \tilde{\mathbf{Y}}\hat{\mathbf{A}}^{\mathrm{T}}\hat{\mathbf{A}}.\end{aligned} \tag{20}$$

From (20), it can be concluded that the proposed A2SAN first estimates abundance maps by $\hat{\mathbf{A}}$, then extracts endmembers by $\tilde{\mathbf{Y}}\hat{\mathbf{A}}^{\mathrm{T}}$, and finally reconstructs the input image by $\tilde{\mathbf{Y}}\hat{\mathbf{A}}^{\mathrm{T}}\hat{\mathbf{A}}$. Obviously, in the proposed A2SAN, we exploit the relationship between the attention mechanism and hyperspectral unmixing and directly use the attention mechanism for hyperspectral unmixing and reconstruction purposes (instead of just using it as a processing module). Experimental results in Section IV show its excellent performance in extracting endmembers, estimating abundance maps, and reconstructing images.

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D. Loss Function

Our proposed A2SN and A2SAN are both end-to-end models and employ abundance-guided mechanisms for final spectral unmixing. Therefore, reconstruction and abundance losses are included in our networks to obtain an accurate reconstructed image and abundance matrix, respectively. The reconstruction loss is computed by

$$L_R = \sum_{i=1}^{Np} \arccos \frac{\hat{\mathbf{y}}_i \cdot \mathbf{y}_i}{||\hat{\mathbf{y}}_i||_2 ||\mathbf{y}_i||_2} \tag{21}$$

where $\hat{\mathbf{y}}_i$ denotes the reconstructed pixel and $\mathbf{y}_i$ is the real pixel. The abundance loss includes two items for meeting the ASC and ANC constraints, respectively. We first compute the ASC loss by

$$L_{\text{ASC}} = \frac{1}{Np} \sum_{j=1}^{Np} \left[\left(\sum_{i=1}^{c} \hat{a}_{ij} \right) - 1 \right]^2. \tag{22}$$

Then, we change the abundance matrix into

$$\hat{a}_{ij} = \begin{cases} 0, & \hat{a}_{ij} > 0 \\ -\hat{a}_{ij}, & \hat{a}_{ij} \leqslant 0 \end{cases} \tag{23}$$

and get the ANC loss by

$$L_{\text{ANC}} = \frac{1}{c \times Np} \sum_{i=1}^{c} \sum_{j=1}^{Np} \hat{a}_{ij} \tag{24}$$

where $\hat{a}_{ij}$ denotes the abundance of the ith endmember on the jth pixel. The final abundance loss is computed by

$$L_A = L_{\text{ASC}} + L_{\text{ANC}}. \tag{25}$$

Finally, the total loss for training the proposed A2SN and A2SAN is defined as

$$L = L_R + \beta L_A \tag{26}$$

where β is a regularization parameter controlling the balance between the reconstruction loss and the abundance loss.

IV. Experimental Results

In this section, the performance of the proposed A2SN and A2SAN is evaluated on a synthetic dataset and three real hyperspectral datasets. In addition, seven classical and state-of-the-art unmixing methods are used for comparison: MVSA [22], MVES [32], orthogonal projection endmember and orthogonal projection abundance (OPE-OPA) [9], uDAS [43], CAE [47], deep autoencoder unmixing (DAEU) [61], multitask autoencoder unmixing (MTAEU) [40], spectral information divergence autoencoder unmixing (SIDAEU) [62], and marginalized denoising autoencoder (mDAE) [63].

A. Dataset Descriptions

1) Synthetic Dataset: The synthetic dataset is selected by using a set of reflectance spectral signatures selected from the United States Geological Survey (USGS) library. The spectral data contains five endmembers and 60 × 60 pixels with 188 spectral bands and a wavelength range from 0.38 to 2.5 μm, with a spectral resolution of 10 nm. In addition, Gaussian noise with an SNR set to 10, 20, and 30 dB is added to the synthetic dataset, respectively.

2) Samson Dataset: The Samson dataset contains 952 × 952 pixels and 156 bands covering the wavelength range from 401 to 889 nm. To reduce the computational cost, we start from the (252, 332)th pixel and use a region of 95 × 95 pixels around it to generate a subcube for our experiments. The final Samson dataset includes three endmembers, that is, soil, tree, and water.

3) Jasper Dataset: The Jasper dataset has 512 × 614 pixels and 224 bands covering the wavelength range from 380 to 2500 nm. We start from the (105, 269)th pixel and consider a region of 100 × 100 around it to generate a subcube for our experiments. Due to dense water vapor and atmospheric effects, we remove the bands 1–3, 108–112, 154–166, and 220–224 and keep 198 bands. The final Jasper dataset contains four endmembers, that is, water, soil, trees, and roads.

4) Urban Dataset: The Urban dataset includes 307 × 307 pixels and 210 bands covering the wavelength range from 400 to 2500 nm. We remove bands 1–4, 76, 87, 101–111, 136–153, and 198–210 (and retain 162 bands) due to dense water vapor and atmospheric effects. The final Urban dataset has four endmembers, that is, asphalt, grass, tree, and roof.

B. Evaluation Metrics

We employ spectral angle distance (SAD) and RE to evaluate the performance of different algorithms for extracting endmembers and reconstructing images. These metrics are, respectively, defined as

$$\text{SAD}(\hat{\mathbf{m}}_i, \mathbf{m}_i) = \arccos \frac{\hat{\mathbf{m}}_i \cdot \mathbf{m}_i}{||\hat{\mathbf{m}}_i||_2 ||\mathbf{m}_i||_2} \tag{27}$$

where $\hat{\mathbf{m}}_i$ and $\mathbf{m}_i$ are the ith estimated and real endmembers, respectively,

$$\text{RE}(\hat{\mathbf{y}}_i, \mathbf{y}_i) = \frac{1}{Np} \sum_{i=1}^{Np} \arccos \frac{\hat{\mathbf{y}}_i \cdot \mathbf{y}_i}{||\hat{\mathbf{y}}_i||_2 ||\mathbf{y}_i||_2} \tag{28}$$

where Np denotes the number of pixels, $\hat{\mathbf{y}}_i$ denotes the ith estimated hyperspectral pixel, and $\mathbf{y}_i$ represents the ith real hyperspectral pixel. The root mean square error (RMSE) is used to evaluate the performance of different algorithms in abundance estimation, and it is computed by

$$\text{RMSE}(\hat{\mathbf{a}}_i, \mathbf{a}_i) = \sqrt{\frac{1}{Np} \sum_{j=1}^{Np} ||\hat{\mathbf{a}}_i - \mathbf{a}_i||_2} \tag{29}$$

where $\hat{\mathbf{a}}_i$ and $\mathbf{a}_i$ denote the abundance of the ith estimated and real endmember over all pixels, respectively.

C. Experiments on the Synthetic Dataset

Tables I and II present the mean SAD and RMSE results obtained by different unmixing methods, and the best and second-best results are in bold and underlined, respectively. From Tables I and II, we conclude that MVSA obtains more accurate endmember extraction results than A2SN on synthetic data with 30 dB noise, and it also has the best abundance estimation results on synthetic data with 30 dB noise, but the

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TABLE I

MEAN SAD OBTAINED BY DIFFERENT UNMIXING METHODS ON SYNTHETIC DATA UNDER DIFFERENT LEVELS OF NOISE. BEST RESULTS ARE IN BOLD AND SECOND-BEST RESULTS ARE UNDERLINED

SNR	MVSA [22]	MVES [32]	CAE [47]	DAEU [61]	MTAEU [40]	SIDAEU [62]	mDAE [63]	A2SN	A2SAN
10dB	0.1620	0.1507	0.4095	0.1277	0.1332	0.1179	0.1441	0.0932	**0.0747**
20dB	0.1870	0.1783	0.3271	0.1430	0.1094	0.2630	0.1275	**0.0529**	0.0912
30dB	0.0469	0.1140	0.2510	0.0796	0.0938	0.1223	0.0574	0.0585	**0.0418**

TABLE II

RMSE OBTAINED BY DIFFERENT UNMIXING METHODS ON SYNTHETIC DATA UNDER DIFFERENT LEVELS OF NOISE. BEST RESULTS IN ARE BOLD AND SECOND-BEST RESULTS ARE UNDERLINED

SNR	MVSA [22]	MVES [32]	CAE [47]	DAEU [61]	MTAEU [40]	SIDAEU [62]	mDAE [63]	A2SN	A2SAN
10dB	0.2367	0.2279	0.3053	0.2513	0.2154	0.1965	0.3204	0.1681	**0.1530**
20dB	0.1765	0.1719	0.2836	0.2174	0.1372	0.2211	0.3235	0.1313	**0.1051**
30dB	**0.1070**	0.1625	0.2657	0.2170	0.2320	0.2340	0.3265	0.1255	0.1098

TABLE III

RE OBTAINED BY DIFFERENT UNMIXING METHODS ON SYNTHETIC DATA UNDER DIFFERENT LEVELS OF NOISE. BEST RESULTS ARE IN BOLD AND SECOND-BEST RESULTS ARE UNDERLINED

SNR	Data Type	MVSA [22]	MVES [32]	CAE [47]	DAEU [61]	MTAEU [40]	SIDAEU [62]	mDAE [63]	A2SN	A2SAN
10dB	Noise Data	0.4137	0.4025	0.3802	0.3592	0.3577	0.3956	0.4258	0.3543	0.3547
	Clear Data	0.2144	0.1898	0.1343	0.0743	0.0579	0.1745	0.2351	**0.0317**	0.0356
20dB	Noise Data	0.1617	0.1349	0.1785	0.1286	0.1219	0.1276	0.1765	0.1239	0.1256
	Clear Data	0.1044	0.0876	0.1265	0.0462	**0.0289**	0.0478	0.1266	0.0351	0.0398
30dB	Noise Data	0.1785	0.1469	0.1831	0.0419	0.0412	0.0470	0.1869	0.0407	0.0346
	Clear Data	0.1741	0.1414	0.1774	0.0230	0.0216	0.0311	0.1828	0.0215	**0.0072**

TABLE IV

PERFORMANCE EVALUATION OF DIFFERENT UNMIXING METHODS ON THE SAMSON DATASET. BEST RESULTS ARE IN BOLD AND SECOND-BEST RESULTS ARE UNDERLINED

Methods		MVSA [22]	MVES [32]	OPE-OPA [9]	uDAS [43]	CAE [47]	DAEU [61]	MTAEU [40]	SIDAEU [62]	mDAE [63]	A2SN	A2SAN
SAD	Soil	0.0455	0.0455	0.0501	0.1140	1.0500	0.0271	0.0393	0.0236	0.3958	0.0243	**0.0208**
	Tree	0.0247	0.0481	**0.0203**	0.1120	0.1614	0.0322	0.0430	0.0243	0.1263	0.0224	0.0259
	Water	0.3724	0.2974	0.0721	0.1787	0.3227	0.0467	**0.0439**	0.0542	0.1906	0.0551	0.0625
Mean SAD		0.1475	0.1303	0.0475	0.1349	0.5114	0.0353	0.0421	0.0340	0.2376	**0.0339**	0.0364
RMSE		0.2747	0.3214	0.2151	0.2884	0.2462	0.0700	0.0786	0.0614	0.3390	0.0469	**0.0443**
RE		0.4578	0.4881	0.2533	0.3896	0.0965	0.0501	0.0544	0.0590	0.1053	**0.0370**	0.0390
Time (s)		0.8615	1.0020	**0.1681**	108.6836	37.4707	43.8835	85.9920	41.2642	11.2525	41.8415	32.6343

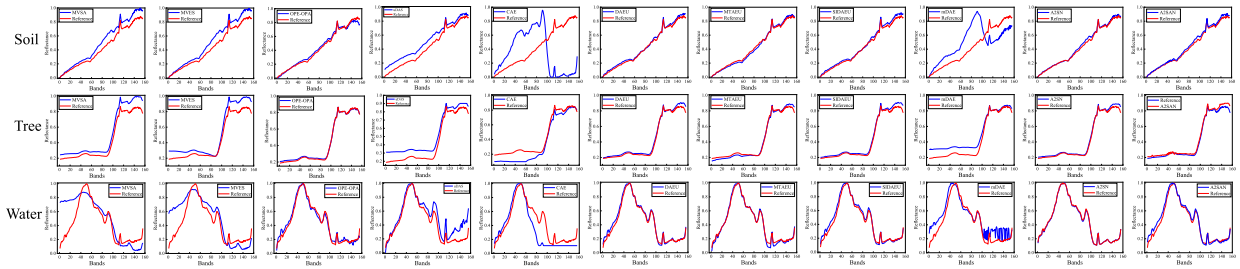


Fig. 6. Visual comparison of the endmembers extracted by different unmixing methods on the Samson dataset.

gap between it and the proposed A2SN and A2SAN is small. Moreover, our proposed A2SN and A2SAN achieved accurate results on endmember extraction and abundance estimation in terms of the values of mean SAD and RMSE compared with other unmixing methods on very noisy synthetic data (10 and 20 dB SNR). Therefore, in terms of the overall endmember extraction and abundance estimation, the proposed A2SN and A2SAN appear to be more effective than

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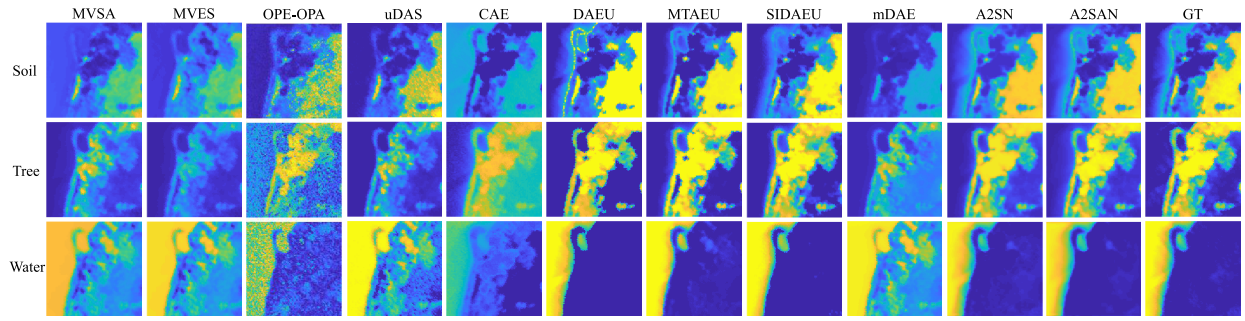


Fig. 7. Visual comparison of the abundance maps estimated by different unmixing methods on the Samson dataset.

TABLE V
PERFORMANCE EVALUATION OF DIFFERENT UNMIXING METHODS ON THE JASPER DATASET. BEST RESULTS ARE IN BOLD AND SECOND-BEST RESULTS ARE UNDERLINED

Methods		MVSA [22]	MVES [32]	OPE-OPA [9]	uDAS [43]	CAE [47]	DAEU [61]	MTAEU [40]	SIDAEU [62]	mDAE [63]	A2SN	A2SAN
SAD	Tree	0.0519	0.0513	0.0650	0.0331	0.2314	**0.0185**	0.0961	0.0300	0.1322	0.0240	0.0297
	Water	0.1309	0.1324	0.3070	0.0292	0.2400	**0.0098**	0.0955	0.0261	0.1810	0.0120	0.0201
	Soil	0.1186	0.1273	0.0784	0.1085	0.2434	0.0506	0.1048	0.0796	0.4673	**0.0390**	0.0502
	Road	0.0601	0.0644	0.0977	0.0532	0.4382	0.2056	0.2172	0.2433	0.1587	0.0934	**0.0428**
Mean SAD		0.0904	0.0939	0.1371	0.0560	0.2883	0.0711	0.1284	0.0948	0.2348	0.0421	**0.0357**
RMSE		0.2094	0.2063	0.2029	0.1461	0.2468	0.1319	0.2700	0.1309	0.2748	**0.0956**	0.1164
RE		0.4406	0.4295	0.4410	0.3683	0.0957	0.0768	0.0888	0.0718	0.1048	0.0700	**0.0652**
Time (s)		0.9601	1.8972	**0.2154**	182.4129	53.3654	47.9196	99.3320	10.7187	12.4647	24.9551	27.5029

other techniques. In particular, MVSA and MVES cannot reconstruct images, and we use their results of endmember extraction and abundance estimation to obtain the corresponding reconstruction results for evaluation. Table III shows the performance of different unmixing methods in the task of reconstructing images. Since we first generate the synthetic data without any noise (clear data) and then add noise to obtain the final synthetic data (noise data), we compare the reconstructed results with both noise data and clear data to test their noise immunity, and the input of all techniques is the noise data. From Table III, it is not difficult to find that the results of A2SN and A2SAN are closer to the clear data. Moreover, although A2SN and A2SAN do not obtain the best results on synthetic data with 20-dB noise, the gap between them and MTAEU is small. This fact suggests that the proposed A2SN and A2SAN not only effectively reconstruct images, but also exhibit more robustness to noise when compared with other techniques.

D. Experiments on the Samson Dataset

Table IV presents the quantitative results obtained by different unmixing methods on the Samson dataset. From Table IV, we observe that the proposed A2SN and A2SAN obtain the best results in the task of reconstructing images and estimating abundance maps, while they differ somewhat from other methods, such as MVSA, MVES, OPE-OPA, uDAS, and mDAE. This observation highlights the significant advantages of the proposed A2SN and A2SAN methods over other unmixing techniques in accurately estimating abundance maps and reconstructing images. In addition, MTAEU exhibits better results in extracting the endmember "water" than other methods, the endmember "tree" can be extracted more accurately by OPE-OPA, and the proposed A2SAN can obtain the best results in extracting the endmember "soil." In summary, in terms of mean SAD, A2SN is the best method, and A2SAN is better than other methods except for SIDAEU and DAEU, but the difference is very small. This fact reveals that both A2SN and A2SAN are very effective in extracting endmembers. Table IV also reports the efficiency of different unmixing methods. From Table IV, we see that OPE-OPA and MVSA are very efficient but they cannot reconstruct images. Therefore, considering that the proposed A2SN and A2SAN have accurate results in extracting endmembers, estimating abundances, and reconstructing images, both A2SN and A2SAN are effective and efficient unmixing methods.

We present the visual results of endmember extraction and abundance estimation obtained by different unmixing methods in Figs. 6 and 7, respectively. Clearly, upon observing Fig. 6, it can be seen that uDAS, CAE, and mDAE exhibit the poorest performance in extracting soil and tree endmembers. Similarly, MVSA, MVES, uDAS, CAE, and mDAE do not yield satisfactory results in extracting water endmembers. On the other hand, the endmembers extracted by A2SN and A2SAN demonstrate closer proximity to the reference endmembers, showcasing their superior performance in overall endmember extraction. From Fig. 7, by closer examination, it is evident that MVSA, MVES, CAE, and mDAE do not achieve satisfactory agreement with the ground truth. However, the proposed A2SN and A2SAN methods showcase remarkable performance in accurately estimating the abundances of each mineral at every pixel, emerging as the top-performing technique in this regard.

E. Experiments on the Jasper Dataset

Table V reports the results of extracting endmembers, estimating abundance maps, and reconstructing images obtained

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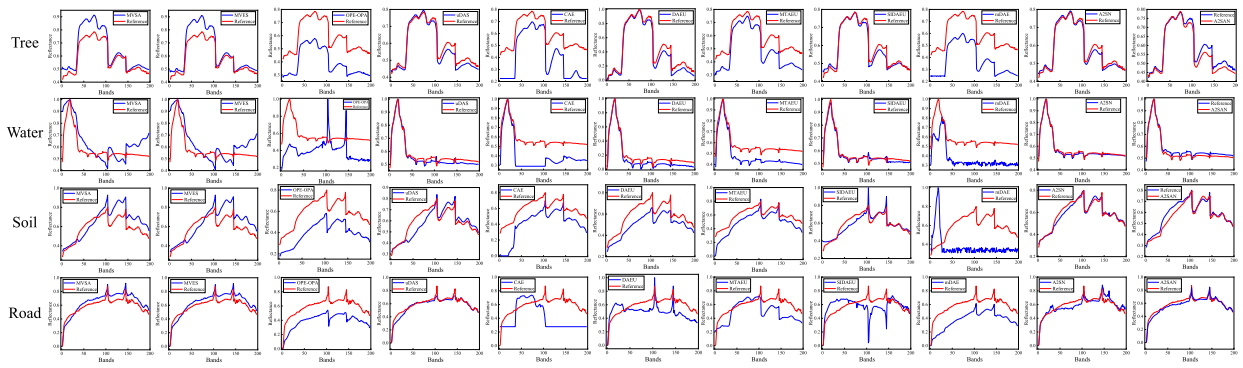


Fig. 8. Visual comparison of the endmembers extracted by different unmixing methods on the Jasper dataset.

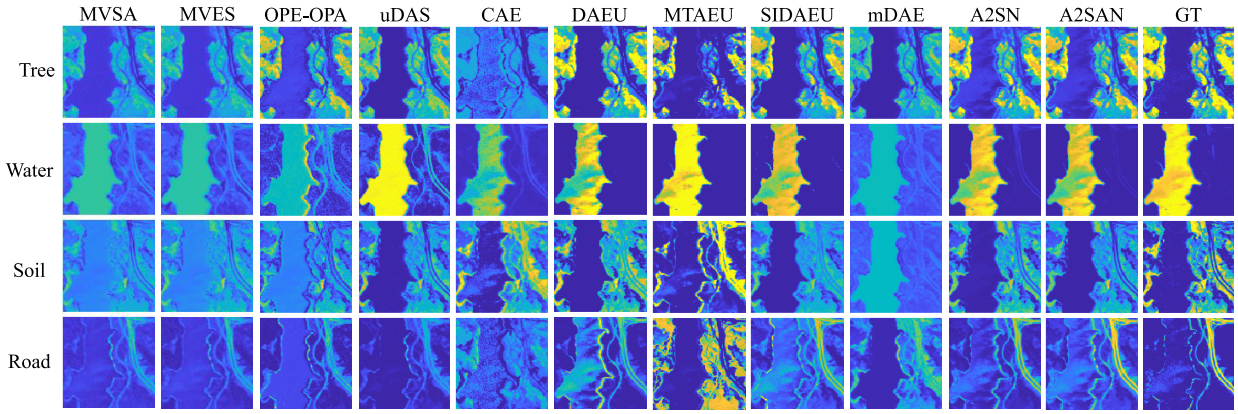


Fig. 9. Visual comparison of the abundance maps estimated by different unmixing methods on the Jasper dataset.

TABLE VI

PERFORMANCE EVALUATION OF DIFFERENT UNMIXING METHODS ON THE URBAN DATASET. BEST RESULTS ARE IN BOLD AND SECOND-BEST RESULTS ARE UNDERLINED

Methods		MVSA [22]	MVES [32]	OPE-OPA [9]	uDAS [43]	CAE [47]	DAEU [61]	MTAEU [40]	SIDAEU [62]	mDAE [63]	A2SN	A2SAN
SAD	Asphalt	0.1585	0.1928	0.2007	0.2135	0.3523	0.2106	0.1411	0.2901	0.2163	**0.0735**	0.1113
	Grass	1.0727	1.0979	0.8057	0.9575	0.9602	0.2171	0.3427	0.3471	0.4843	0.0520	**0.0494**
	Tree	0.0946	0.1116	0.1602	0.1110	0.1629	0.0806	0.0773	0.1338	0.1567	**0.0199**	0.0557
	Roof	0.1299	0.1306	0.2443	0.1317	0.2044	0.2436	0.1473	0.2555	0.2105	**0.0945**	0.1337
Mean SAD		0.3639	0.3832	0.3527	0.3534	0.4199	0.1880	0.1771	0.2566	0.2670	**0.0600**	0.0875
RMSE		0.2939	0.2767	0.2536	0.3127	0.2679	0.1454	0.1560	0.1670	0.2704	**0.1013**	0.1065
RE		0.4295	0.4784	0.2489	0.5325	0.1696	0.0548	0.0713	**0.0509**	0.1151	0.0650	0.0540
Time (s)		2.7708	1.5079	**0.7896**	1731.6	441.3218	52.8121	107.5883	48.4249	103.4156	180.3824	214.4112

by different unmixing techniques on the Jasper dataset. It is easy to find that DAEU gets the best results in extracting the "tree" and "water" endmembers, while A2SN and A2SAN get the best results in extracting the "soil" and "road" endmembers, respectively. In general, A2SN and A2SAN are better than other unmixing techniques considering the mean SAD. Moreover, in terms of RMSE and RE, the proposed A2SN and A2SAN are more effective than other methods in estimating abundance maps and reconstructing images. Table V also reports the running time obtained by different unmixing methods. From Table V, we see that OPE-OPA and MVSA are very fast but they lack the ability to reconstruct images. Therefore, although A2SN and A2SAN may not exhibit the highest efficiency, their performance is still deemed satisfactory for hyperspectral unmixing purposes.

Fig. 8 provides a visual comparison of the endmembers extracted using various unmixing techniques on the Jasper dataset. Upon observation, it is apparent that CAE and mDAE do not yield good results in extracting all minerals, MVSA and MVES struggle to accurately extract the "water" and "soil" endmembers, OPE-OPA does not obtain good results in extracting the "water" endmember, and MTAEU has the best result in extracting the "soil" and "road" endmembers. In contrast, the curves obtained by the proposed A2SN and A2SAN methods demonstrate a remarkable agreement with all minerals, showcasing their superior performance in endmember extraction. Fig. 9 illustrates the abundance maps estimated by various unmixing methods on the Jasper dataset. It is evident that MVSA, MVES, OPE-OPA, uDAS, CAE, and mDAE do not yield good results in abundance estimation.

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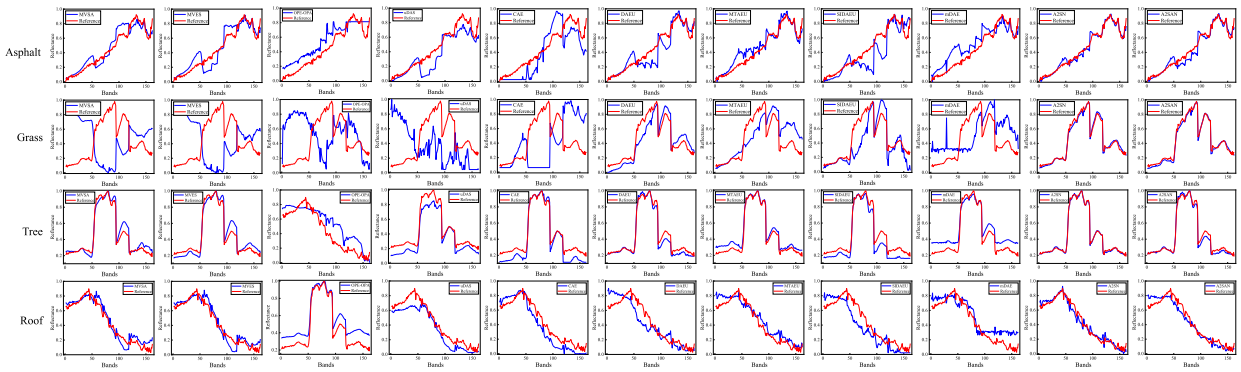


Fig. 10. Visual comparison of the endmembers extracted by different unmixing methods on the Urban dataset.

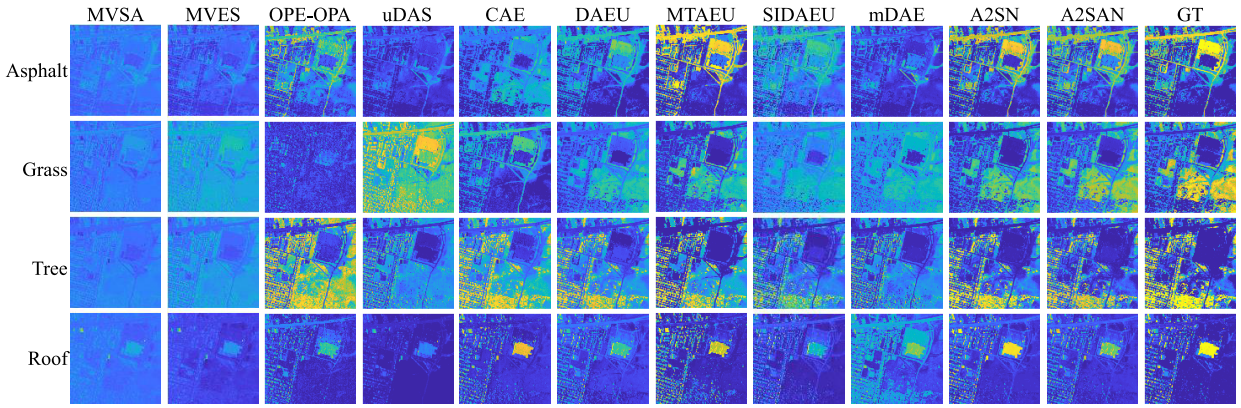


Fig. 11. Visual comparison of the abundance maps estimated by different unmixing methods on the Urban dataset.

In contrast, the abundance maps obtained by the proposed A2SN and A2SAN methods exhibit a closer resemblance to the ground truth, highlighting their effectiveness in accurately estimating abundances.

F. Experiments on the Urban Dataset

Table VI evaluates the performance of different unmixing methods in extracting endmember signatures, estimating abundance maps, and reconstructing images on the Urban dataset. We observe that A2SN obtains the best results in extracting "asphalt," "tree," and "water" endmembers and A2SAN gets more accurate results than other methods in extracting the "grass" endmembers. Generally, A2SN and A2SAN are more effective than other unmixing techniques in terms of endmember extraction, and the difference between them and other methods is quite significant. Moreover, the proposed A2SN and A2SAN obtain more accurate results than other methods in terms of RMSE, and this indicates that they can estimate abundance maps more effectively than other unmixing techniques. Furthermore, SIDAEU obtains the best results in hyperspectral reconstruction, but with little margin with the proposed A2SN and A2SAN. Therefore, A2SN and A2SAN are also competitive in terms of hyperspectral reconstruction. Table VI also reports the running time obtained by different unmixing methods. From Table VI, we see that OPE-OPA and MVES have high efficiency. Although the proposed A2SN and A2SAN do not obtain the best results in the running time, OPE-OPA and MVES cannot directly reconstruct images and A2SA and A2SAN can also be considered effective and efficient unmixing methods.

TABLE VII
PARAMETER SETTINGS OF A2SN AND A2SAN ON DIFFERENT DATASETS

Parameter Settings	A2SN			A2SAN		
	Samson	Jasper	Urban	Samson	Jasper	Urban
Learning Rate	1e-3	5e-2	6e-2	1e-3	5e-2	3e-4
Regularization Parameter	1.5e-6	7.5e-4	1e-5	1.5e-6	1e-4	1e-4
Weight Decay	1e-4	1e-4	1e-4	1e-4	1e-4	1e-6

Fig. 10 depicts the endmember signatures extracted by different unmixing techniques. From Fig. 10, it can be observed that our proposed A2SN and A2SAN exhibit better consistency than other methods, especially when extracting the "grass" endmember. Fig. 11 visually reports the abundance maps obtained by different unmixing techniques on the Urban dataset. From Fig. 11, upon observation, it is evident that most of the methods, except for DAEU, MTAEU, and SIDAEU, yield unsatisfactory results in abundance estimation. Notably, the abundance maps generated by A2SN and A2SAN exhibit the highest degree of similarity to the ground truth, outperforming the other methods in abundance estimation.

G. Parameter Setting and Parameter Analysis

1) Parameter Settings: In our experiments, we empirically set the number of layers and neurons in the spectral and spatial

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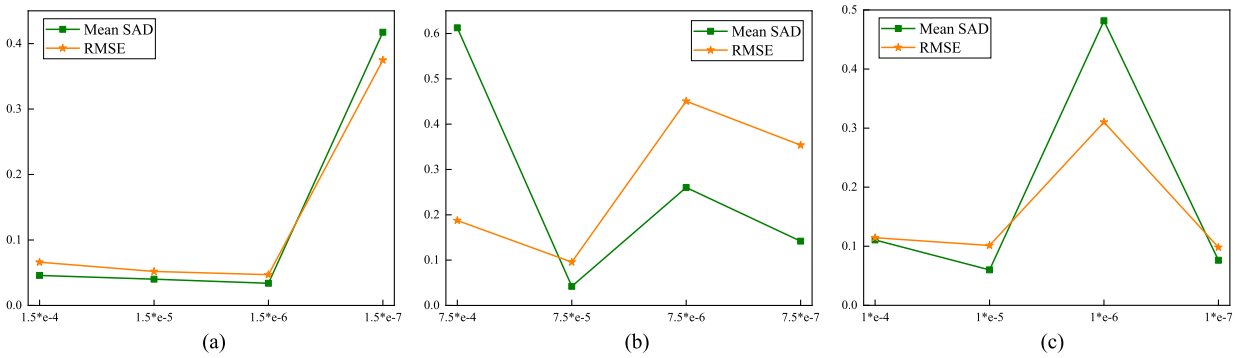


Fig. 12. Parameter analysis of different regularization parameters β for the proposed A2SN on real datasets. (a) Samson. (b) Jasper. (c) Urban.

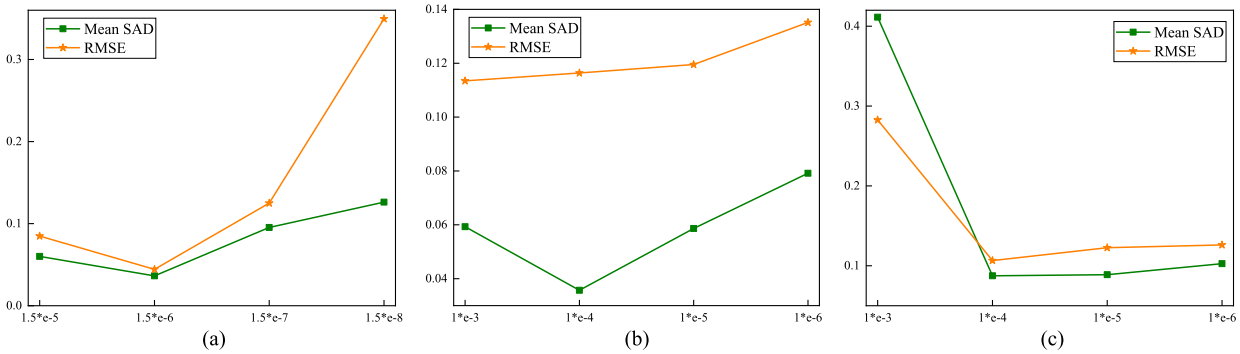


Fig. 13. Parameter analysis of different regularization parameters β for the proposed A2SAN on real datasets. (a) Samson. (b) Jasper. (c) Urban.

modules of our proposed A2SN and A2SAN. In their corresponding unmixing module, we design a new loss function given by (26) to obtain good results in extracting endmember signatures, estimating abundance maps, and reconstructing images. The detailed parameter settings of A2SN and A2SAN have been reported in Table VII.

2) Parameter Analysis: Fig. 12 shows the parameter analysis results of different regularization parameters β for the proposed A2SN on the Samson, Jasper, and Urban datasets. From Fig. 12, we see that A2SN will obtain the best results when the parameters β are $1.5 * e - 6$, $7.5 * e - 5$, and $1 * e - 5$ on Samson, Jasper, and Urban, respectively. Fig. 13 reports the parameter analysis results of different regularization parameters β for the proposed A2SAN on the Samson, Jasper, and Urban datasets. From Fig. 13, we observe that A2SAN will have the most accurate result on the Samson dataset when β is $1.5 * e - 6$, and it is also more effective when β is $1 * e - 4$ on the Jasper and Urban datasets.

TABLE VIII
ABLATION STUDY USING OUR PROPOSED A2SN AND A2SAN ON DIFFERENT DATASETS WITH DIFFERENT MODULES. BEST RESULTS ARE IN BOLD

Datasets	Spectral Module	Spatial Module	A2SN		A2SAN	
			Mean SAD	RMSE	Mean SAD	RMSE
Synthetic (20dB)	✓	✗	0.1735	0.1673	0.1529	0.1494
	✗	✓	0.1594	0.1320	0.1173	0.1240
	✓	✓	**0.0529**	**0.1313**	**0.0912**	**0.1051**
Samson	✓	✗	0.0484	0.0548	0.0490	0.0629
	✗	✓	0.0348	0.0587	0.0563	0.0587
	✓	✓	**0.0339**	**0.0469**	**0.0364**	**0.0443**
Jasper	✓	✗	0.0521	0.1024	0.0433	0.1176
	✗	✓	0.0470	0.1036	0.0402	0.1257
	✓	✓	**0.0421**	**0.0956**	**0.0357**	**0.1164**
Urban	✓	✗	0.0813	0.1074	0.1043	0.1068
	✗	✓	0.0889	0.1139	0.0936	0.1162
	✓	✓	**0.0600**	**0.1013**	**0.0875**	**0.1065**

H. Discussion

1) Ablation Study: Our proposed A2SN and A2SAN incorporate convolution kernels of different sizes to construct spectral and spatial modules. These modules are then combined to simultaneously learn spectral and spatial features for hyperspectral unmixing. The advantage of integrating the two modules is demonstrated in Table VIII, which presents the mean SAD and RMSE results obtained by A2SN and A2SAN. By comparing the results, we can observe that the combination of spectral and spatial modules achieves the best performance in terms of mean SAD and RMSE across all datasets. This finding highlights the effectiveness of the A2SN and A2SAN design for hyperspectral unmixing applications.

2) Role of the Abundance-Guided Mechanism: The proposed A2SN and A2SAN both convolve the concatenated features to obtain the reconstructed image $\tilde{\mathbf{Y}}$ and then connect their corresponding unmixing module to obtain the final reconstructed image $\hat{\mathbf{Y}}$ by (17) and (20), respectively. The main role of the unmixing module in A2SN and A2SAN

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TABLE IX
RE OBTAINED BY A2SN AND A2SAN CONSIDERING DIFFERENT RECONSTRUCTED IMAGES. BEST RESULTS ARE IN BOLD

Methods	Reconstructed image	Synthetic (20dB)	Samson	Jasper	Urban
A2SN	$\tilde{\mathbf{Y}}$	0.3189	0.1790	0.2505	0.2581
	$\hat{\mathbf{Y}}$	**0.0317**	**0.0370**	**0.0700**	**0.0652**
A2SAN	$\tilde{\mathbf{Y}}$	0.4158	0.1372	0.2276	0.3487
	$\hat{\mathbf{Y}}$	**0.0356**	**0.0390**	**0.0650**	**0.0540**

is to provide an abundance-guided mechanism that not only estimates signatures and abundance maps simultaneously, but also weights the reconstructed image $\tilde{\mathbf{Y}}$ to obtain more accurate reconstruction results $\hat{\mathbf{Y}}$. Table IX reports different reconstructed images that are used to evaluate the performance of the abundance-guided process in A2SN and A2SAN. The reconstructed images $\tilde{\mathbf{Y}}$ and $\hat{\mathbf{Y}}$ are obtained by convolving the concatenated features and performing the unmixing module based on such abundance-guided mechanism. By observing Table IX, it is easy to conclude that A2SN and A2SAN considering $\hat{\mathbf{Y}}$ obtain better results on all datasets. This fact indicates that our unmixing module based on an abundance-guided mechanism can not only perform spectral unmixing, but also reconstruct images very effectively.

V. CONCLUSION AND FUTURE WORK

In this article, we propose a new network called A2SN for hyperspectral unmixing. The proposed A2SN employs spectral and spatial modules to simultaneously learn spectral and spatial information and then design an unmixing module that uses an abundance-guided mechanism to simultaneously extract endmembers, estimate abundance maps, and reconstruct images. Furthermore, we also embed an attention mechanism into A2SN to further develop an A2SAN for hyperspectral unmixing. The proposed A2SAN directly uses attention for hyperspectral unmixing rather than just as a module to extract information. Especially, existing DL-based unmixing methods both estimate endmembers and abundances from the weight and output of a specific layer, but our proposed A2SN and A2SAN use a totally different strategy, that is, abundance-guided, to perform spectral unmixing. Moreover, A2SN and A2SAN exhibit robustness to noise because they weight reconstructed images by (17) and (20), respectively. Experimental results show that our newly proposed A2SN and A2SAN exhibit excellent performance in extracting endmembers, estimating abundance maps, and reconstructing images compared with other unmixing methods. In our future work, we will design a module that can not only simultaneously estimate endmember signatures and abundance maps, but also automatically determine the number of endmembers.

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